

Model validation review - GBM and CLAM

Revised: 2026-09-22. Role: self-validation project adopting a second-line review structure; developer and reviewer are the same person. This is not organizationally independent validation or a regulatory approval.

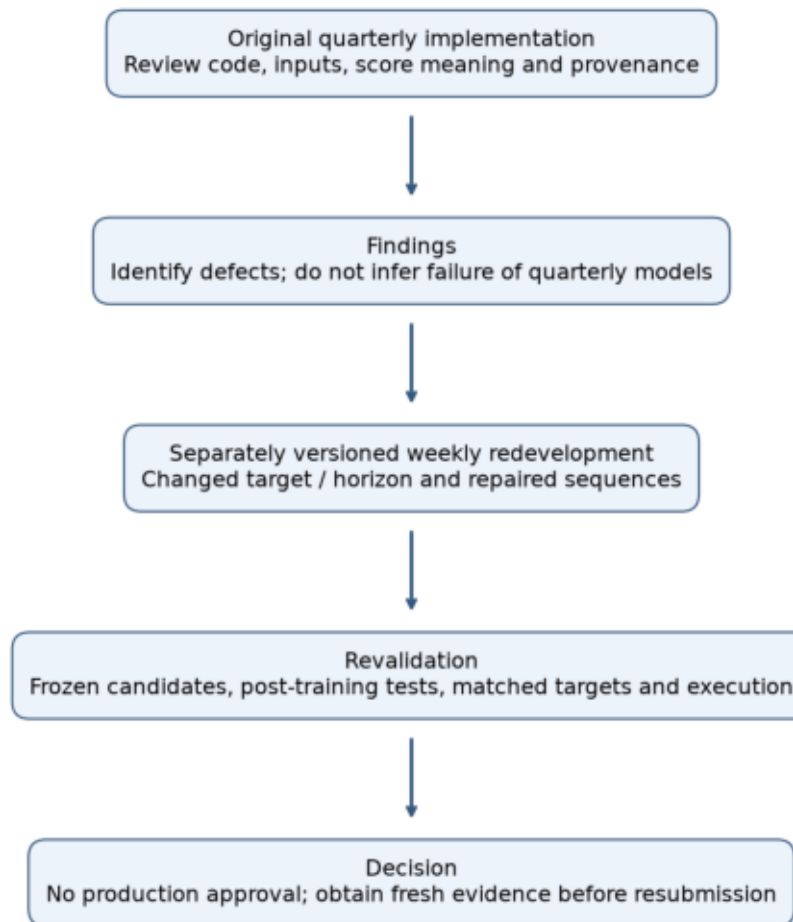
1. Executive decision

No candidate is approved for production. Momentum is retained as a comparison rule, not an approved trading model. These are project decisions under the evidence and limitations below, not proof that the model families have no predictive value.

The story has two distinct stages: review the original quarterly forecasters and their proposed weekly use; then assess separately versioned weekly redevelopments. Repairing an implementation defect and changing the prediction horizon are different interventions. This project does not establish that a properly implemented quarterly strategy cannot work.

The revised OOT table below supersedes earlier report numbers. In particular, whole-sample significance, shortened lagged holding periods and the old CLAM direction metric must not be used as approval evidence. The evaluation ends on 2026-08-28. Q1 uses 241 complete weekly holding periods after 2022-01-03, with entry and exit prices both observed by that cutoff. This is after the recorded training/selection cutoff, but the period has already been inspected in earlier rounds and is not an untouched holdout.

2. Review trail and scope



stage	purpose	evidence / consequence
Original quarterly implementation	Examine original code, input construction, score meaning and provenance	GBM single-path noise; CLAM cross-ticker windows; incorrect direction metric; original-weight training cutoff unverified

stage	purpose	evidence / consequence
Original-methodology twin	Freeze training at 2021-12-31 to examine later observations	clam_2021; retains original sequence and historical metric defects deliberately
Weekly redevelopment	Change intended prediction target/horizon and repair input construction	Separate gbm_weekly / clam_weekly_* model versions, not a successful quarterly validation
Revalidation	Apply shared execution, OOT tests, monitoring and controls	Results below; no production approval
Next approval review	Freeze specifications and seek new evidence	Untouched future data, historical universe data, complete experiments and operational execution evidence

The portfolio application here is weekly top-50 equal-weight long-only selection. It is not a native quarterly-holding backtest. Original-horizon target-matched diagnostics are reported separately in section 7. Risk limits, capacity and live execution are not validated.

3. Model identity and reproduction differences

candidate	identity	fidelity / limitation
gbm	Original 63-step, $T=0.25$, single-path mean-price score	Two-calendar-year rolling window, at least 454 returns; adjusted panel close; includes current signal close; deterministic full-run seed. Controlled reproduction, not verbatim replay of live downloads/RNG
gbm_expected	Closed-form mean-path expectation of that implemented formula	Monotone in the fitted mean log return; analytical diagnostic, not a separately trained model
clam_orig	Historical quarterly weights	Original h5/scaler absent in this checkout; cutoff unverified. File mtime cannot establish training membership
clam_2021	Saved original-methodology twin, cutoff 2021-12-31	Existing weights unchanged; original sequence and metric defects retained for historical comparison
gbm_weekly	Frozen alias gbm_w_126_expected, selected in the earlier development exercise	Six lookback/score variants logged. Alias was not reselected after this review changed execution
clam_weekly_*	Saved per-ticker weekly scalar-target redevelopments	Historical weights unchanged; the universe-size pair is separately trained. Demeaned-return and rank scores are not raw predicted returns
momentum	Fixed 12-1 ranking rule	Comparator under the same snapshot/execution assumptions; no production approval

A 63-step original GBM must not be described as 65 steps. GBM and CLAM quarterly outputs also have different meanings: mean simulated path versus terminal cumulative High log changes. The GBM formula uses the source's mean-log-return drift convention; reproducing that formula is not validation of its economic assumptions. Original forecast error and portfolio selection performance answer different questions.

4. Data and executable backtest contract

The price observation window is 2013-01-02 through 2026-08-28. Later prices retained in the raw archive are excluded from this evaluation. The existing universe snapshot is as of 2026-09-15, so its later membership remains a disclosed source of bias. Re-running a live screener/download is a new dataset, not exact reproduction. The universe contains survivors and was selected using later market capitalizations. Subtracting an equal-weight comparator does not cancel this bias; strategy-specific effects remain unknown. Listing coverage describes the sample and does not measure the size of return bias.

Signals use week-end close information. Orders are assumed filled at the next session close, and held until the following rebalance's next-session close. A common SPY trading calendar determines all dates. Missing ticker quotes are not replaced by the next available future quote. Same-close trades are retained only as optimistic diagnostics (*_same_close), with no claim they were executable.

2026-08-28 is the final observation date. No later price is used for signals, portfolio returns, calibration or the CLAM metric audit. Weekly holding periods and forecast targets that are not complete by the cutoff are excluded. Missing held-name returns within completed periods raise an error. The same fixed cutoff applies to every candidate.

Tradability at signal time: price ≥ 5 , *trailing20 – sessionaveragedollarvolume* $\geq 5M$. Top 50 available scores are equally weighted. Costs are 10 bps per unit of absolute traded weight, including drift in previous holdings; the universe comparator also pays rebalancing costs. This remains a simplified close-fill assumption without capacity/market-impact evidence. Cash earns zero. Sharpe uses zero cash return; active SR is strategy minus equal-weight comparator.

5. Q1 - evidence after training/selection

Stationary bootstrap uses 5,000 draws and mean block 13 weeks; positive 95% CI is the bootstrap screen. The 500-draw permutation comparison uses gross active returns and $p = (1 + \text{exceedances}) / (501)$. It tests exchangeability of names within each scored cross-section, conditional on this biased snapshot; it does not preserve all sector, exposure or serial score structure. Neither test alone is proof of deployable skill.

model	active SR [95% CI]	bootstrap	permutation p	permutation	DSR (provisional)
momentum	0.90 [0.26, 1.53]	PASS	0.020	PASS	0.390
gbm	0.06 [-0.80, 0.86]	FAIL	0.084	FAIL	0.019
gbm_expected	0.39 [-0.31, 1.12]	FAIL	0.164	FAIL	0.086
gbm_weekly	0.50 [-0.12, 1.15]	FAIL	0.106	FAIL	0.129
clam_2021	0.36 [-0.37, 1.05]	FAIL	0.136	FAIL	0.075
clam_weekly_cs_demeaned	-0.89 [-1.56, -0.19]	FAIL	0.838	FAIL	0.000
clam_weekly_cs_rank_small	-0.09 [-0.82, 0.65]	FAIL	0.323	FAIL	0.008

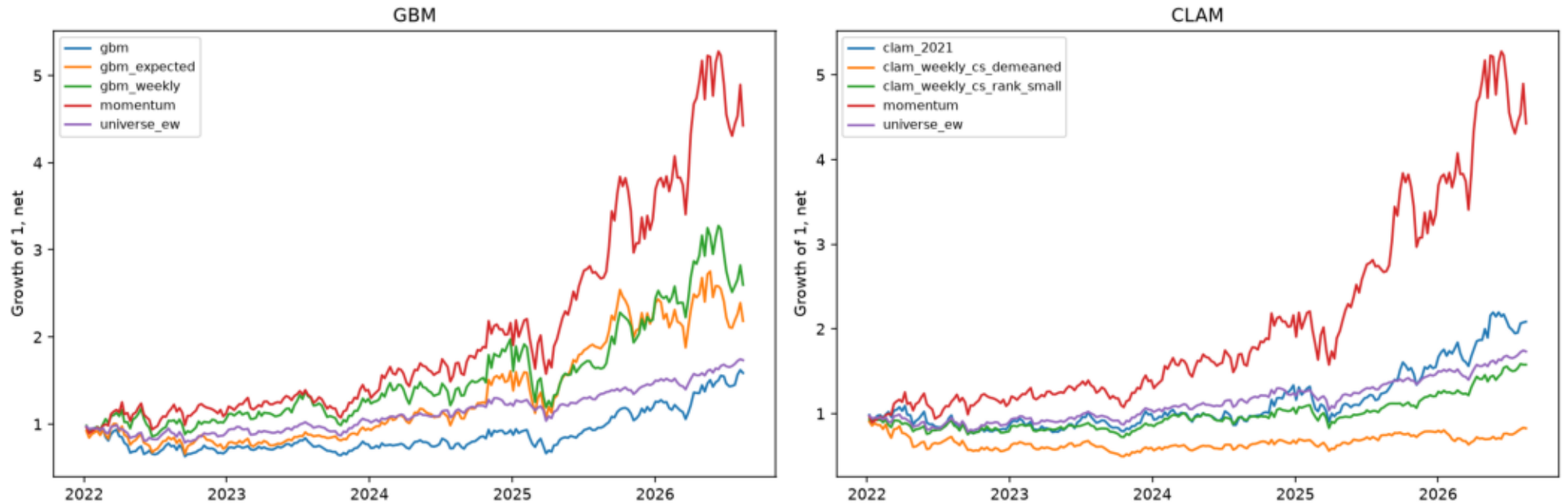
model	active SR [95% CI]	bootstrap	permutation p	permutation	DSR (provisional)
clam_weekly_cs_rank_small_n94_seed20260922	0.09 [-0.83, 1.01]	FAIL	0.307	FAIL	0.022
clam_weekly_cs_rank_small_n500_seed20260922	-0.04 [-0.80, 0.75]	FAIL	0.267	FAIL	0.011
clam_weekly_cs_rank_small_n3000_seed20260922	20.99 [-1.65, -0.35]	FAIL	0.902	FAIL	0.000

DSR is provisional, not PASS/FAIL approval evidence. experiments.json records a minimum of 17 trials: historical baseline/development runs plus the two predeclared training-universe experiments. The gbm_weekly alias is not counted twice. 15 trial return series are available: raw-target and collapsed weekly-bar CLAM runs lack recoverable scores. The variance estimate uses the 15 available trial Sharpes on the same OOT window and N=17; unknown prior searches and correlation between trials remain limitations. The historical search count must never be reduced to obtain approval.

6. OOT portfolio outcomes and controls

model	weeks	CAGR	vol	Sharpe	max DD	active SR
momentum	241	37.8%	41.1%	0.987	-28.6%	0.898
gbm	241	10.5%	31.5%	0.474	-37.1%	0.065
gbm_expected	241	18.4%	41.3%	0.614	-34.5%	0.391
gbm_weekly	241	22.9%	41.2%	0.706	-39.6%	0.501
clam_2021	241	17.2%	34.2%	0.635	-32.4%	0.360
clam_weekly_cs_demeaned	241	-4.1%	29.7%	0.010	-50.6%	-0.895
clam_weekly_cs_rank_small	241	10.3%	24.7%	0.522	-27.8%	-0.089
clam_weekly_cs_rank_small_n94_seed20260922	241	9.2%	37.4%	0.423	-49.8%	0.087
clam_weekly_cs_rank_small_n500_seed20260922	241	11.0%	24.2%	0.552	-32.8%	-0.040
clam_weekly_cs_rank_small_n3000_seed20260922	241	-0.3%	15.3%	0.058	-25.5%	-0.988
universe_ew	241	12.6%	19.9%	0.696	-21.3%	n/a
benchmark_spy	241	12.7%	16.7%	0.802	-21.8%	-0.054

Frozen candidates, revised OOT execution (not an untouched holdout)

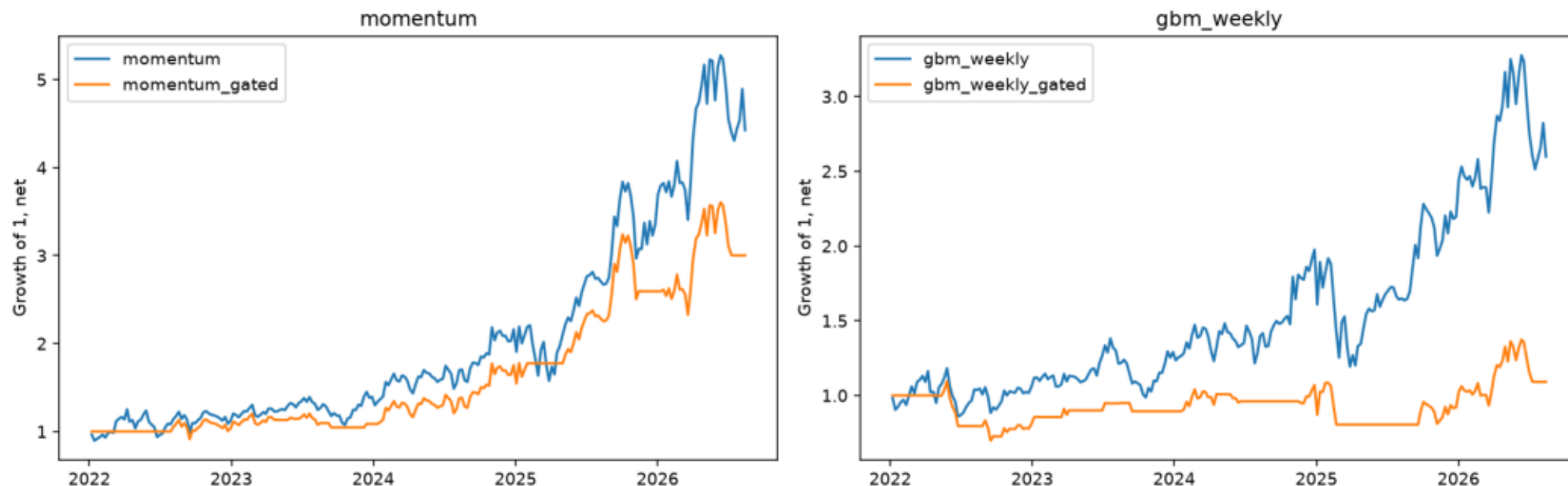


Point estimates describe the tested implementations and this snapshot. Failure to reject a null does not establish absence of skill. The four CLAM specifications do not establish that OHLCV or sequence models in general cannot work. Benchmark-relative approval requires fresh evidence of incremental value, not only a favorable absolute CAGR.

model	CAGR ungated / gated	max DD ungated / gated	Sharpe ungated / gated	off
momentum	37.8% / 26.8%	-28.6% / -28.2%	0.99 / 0.91	33.6%
gbm	10.5% / 0.0%	-37.1% / 0.0%	0.47 / n/a	100.0%
gbm_expected	18.4% / 3.5%	-34.5% / -21.5%	0.61 / 0.30	85.9%
gbm_weekly	22.9% / 1.9%	-39.6% / -36.3%	0.71 / 0.21	61.0%
clam_2021	17.2% / 0.0%	-32.4% / 0.0%	0.63 / n/a	100.0%
clam_weekly_cs_demeaned	-4.1% / 0.0%	-50.6% / 0.0%	0.01 / n/a	100.0%
clam_weekly_cs_rank_small	10.3% / 0.0%	-27.8% / 0.0%	0.52 / n/a	100.0%

model	CAGR ungated / gated	max DD ungated / gated	Sharpe ungated / gated	off
clam_weekly_cs_rank_small_n94_seed20260922	9.2% / 0.0%	-49.8% / 0.0%	0.42 / n/a	100.0%
clam_weekly_cs_rank_small_n500_seed20260922	11.0% / 0.0%	-32.8% / 0.0%	0.55 / n/a	100.0%
clam_weekly_cs_rank_small_n3000_seed20260922	-0.3% / 0.0%	-25.5% / 0.0%	0.06 / n/a	100.0%

Ungated / gated: next-session-close execution, OOT

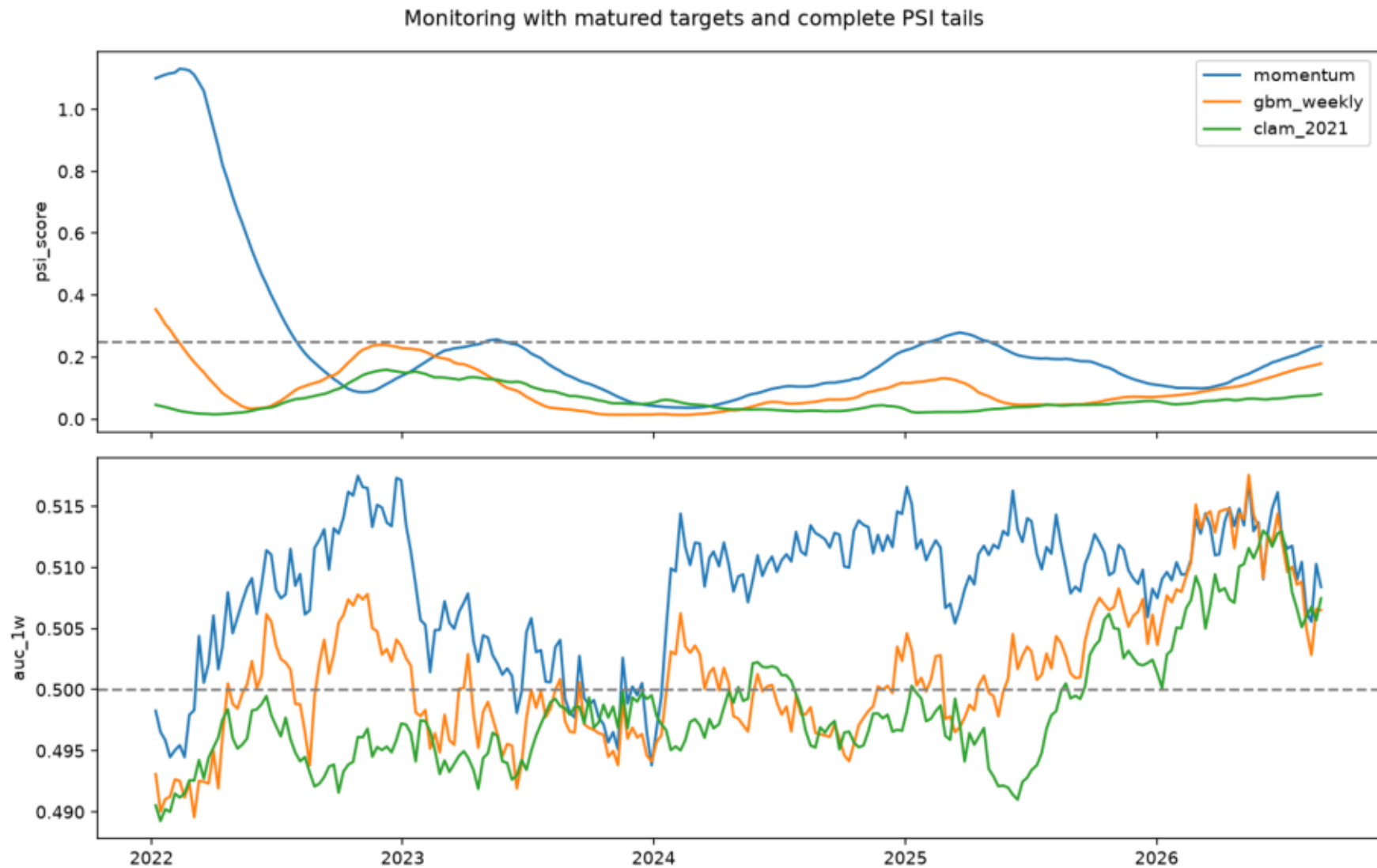


The gate uses $\text{PSI} > 0.25$, trailing active $\text{SR} < 0$, $\text{AUC} < 0.50$ or relative drawdown $< -15\%$. Missing required metrics switch the strategy off. Portfolio metrics are delayed two signal weeks because the preceding delayed-execution holding period has not ended at the current signal close. Entry/exit/rebalance costs are recomputed from actual gated holdings, not added to hypothetical ungated turnover. Relative drawdown uses the ratio of strategy and benchmark wealth.

Thresholds are documented conventions, not empirically established universal cutoffs. The historical record does not establish prospective preregistration. Reuse of the OOT period and any further changes to thresholds require fresh evaluation. The gate remains an experimental control, not the sole approved risk control.

7. Monitoring, targets and CLAM metric audit

model	psi_score	psi_input	auc_1w	rolling_sharpe	active_drawdown
momentum	0.236	0.614	0.508	0.903	-0.217
gbm	0.129	0.614	0.502	1.361	-0.292
gbm_expected	0.183	0.614	0.505	-0.004	-0.316
gbm_weekly	0.179	0.614	0.507	0.826	-0.257
clam_2021	0.080	0.614	0.507	1.247	-0.299
clam_weekly_cs_demeaned	0.148	0.614	0.498	-0.911	-0.702
clam_weekly_cs_rank_small	0.125	0.614	0.502	1.419	-0.329
clam_weekly_cs_rank_small_n94_seed20260922	0.407	0.614	0.505	0.964	-0.268
clam_weekly_cs_rank_small_n500_seed20260922	0.133	0.614	0.508	0.897	-0.270
clam_weekly_cs_rank_small_n3000_seed20260922	0.092	0.614	0.509	-1.739	-0.643



PSI bins include infinite tails, so drifted observations are counted. The reference is the first 104 signal weeks; PSI is not reported before that reference exists. Next-execution-period AUC uses only completed targets; 65-session diagnostics use actual target maturity dates. AUC is pooled across stocks/dates and is not a top-50 portfolio skill test.

Calibration below pairs each available raw-return score with its own target and horizon, uses OOT observations matured by 2026-08-28, and reports

descriptive pooled coefficients without significance claims. In particular a weekly score's slope against a 13-week return is only an association, never evidence of calibration.

model	horizon_sessions	n	slope	intercept	mae
gbm	63	470585	-0.004	0.018	0.137
gbm_expected	63	470585	-0.074	0.019	0.093
gbm_weekly	5	518297	-0.041	0.003	0.044
clam_2021	65	477325	0.002	0.032	0.368

GBM targets the mean of adjusted prices from t through t+63 relative to t; weekly GBM targets adjusted Close[t+5]/Close[t]-1; original CLAM targets raw High[t+65]/High[t]-1. Rank/de-meanned weekly CLAM targets lack the exact training cross-section mapping required for comparable return calibration, so no calibration claim is made for them. Overlapping outcomes and stock dependence limit interpretation.

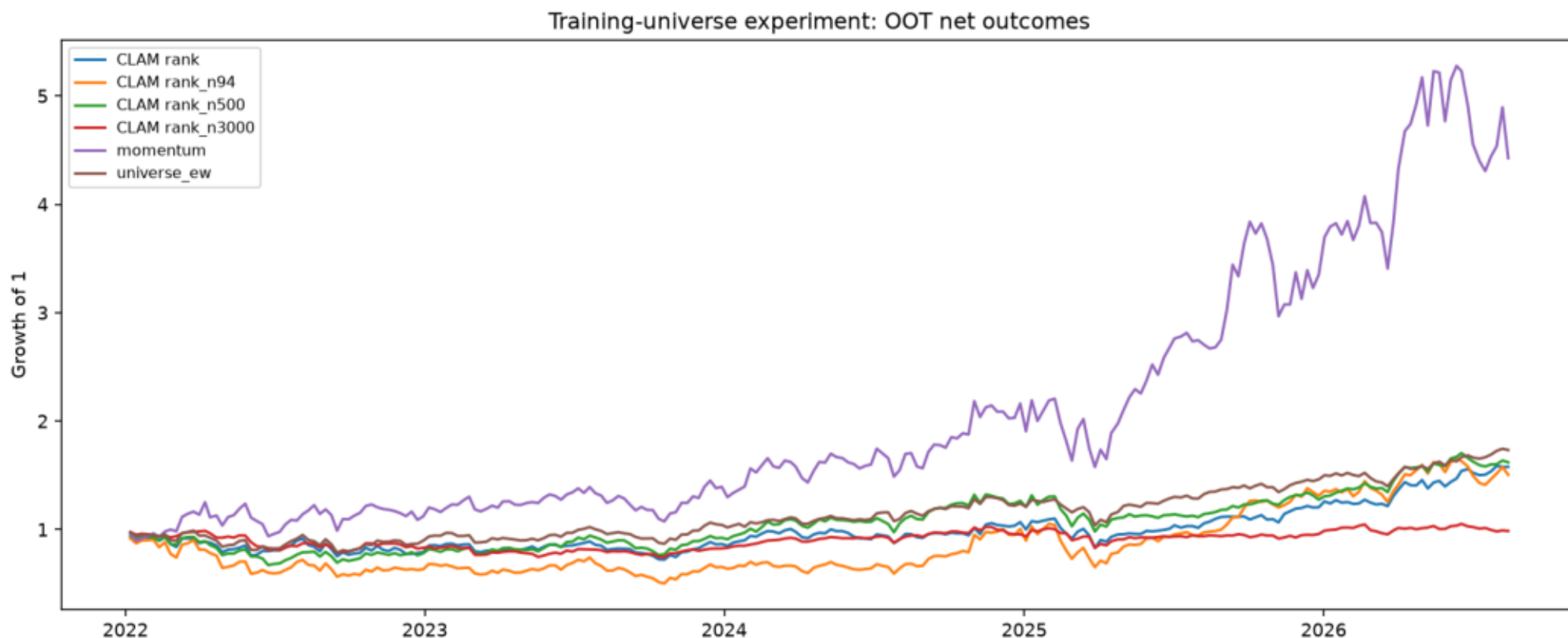
The frozen 2021 twin was evaluated on 1478 single-stock OOT windows (96070 daily Close targets). On the SAME predictions, legacy scaled-sign accuracy was 59.08% and inverse-scaled return-direction accuracy was 49.79%.

This fixed alphabetical sample is a diagnostic of the metric, not a reconstruction of the historical 76%/64% training/validation result. MinMax scaling to (-1,1) does not generally map a zero return to zero; the corrected metric compares scaled values against the scaler's image of zero. New original-model training defaults to this corrected metric and early-stopping monitor. Existing weights were not retrained. The historical twin script explicitly requests the legacy metric for fidelity. Cross-ticker sequences remain a separate confirmed defect; the claim that the network specifically learned same-day market direction is withdrawn.

7.1 Training-universe size: matched 94 / 500 / 3000 experiment

The historical 94-name training list reflected compute constraints, so this experiment varies only training-universe size on the frozen small weekly rank-target architecture. All three runs share seed 20260922, callbacks and a purged validation boundary, and select tickers by market-cap rank; none was selected using OOT outcomes. Training targets end before 2019-12-31, model-selection targets by 2021-12-31. Requested size differs from usable tickers because later listings lack training history. A smaller universe also means a smaller ranking peer group, so validation rank IC is not comparable across rungs on equal terms.

requested_tickers	usable_tickers	training_windows	epochs	validation_rank_ic	oot_cagr	oot_active_sharpe	bootstrap	permutation_p
94	89	27869	8	0.067	9.2%	0.087	FAIL	0.307
500	476	142185	7	0.024	11.0%	-0.040	FAIL	0.267
3000	2567	659736	9	0.016	-0.3%	-0.988	FAIL	0.902



Paired annualized mean net-return differences against the matched 500-name control: 94 2.48% [-14.56%, 21.78%]; 3000 -12.46% [-23.06%, -1.20%]. These are mean-return differences, not CAGR differences. Single-seed evidence on a reused historical period cannot establish a general size effect. The original 500-name artifact is a historical comparator; the newly trained 500-name run is the matched control. A smaller training universe also changes the rank-target peer group, so size and task difficulty are not separated here. Existing production non-approval remains: fresh holdout, universe-bias and operational evidence are still required.

8. Findings, remediation and closure evidence

ID	finding / cause	remediation / evidence	status
F1	Single MC path adds ranking noise	Analytical expectation comparator; revised OOT tables	Original implementation not approved; correction does not prove alpha
F2	Expected GBM rank is monotone in mean log return	Explicit model identity and benchmark comparison	Documented; economic rationale still required

ID	finding / cause	remediation / evidence	status
F3	Original CLAM mixes tickers within windows	Source review; weekly redevelopment uses per-ticker sequences	Original defect open; repaired structure is a separate model
F4	Original weights' training provenance unverified	mtime inference removed; original artifacts absent; twin cutoff separately recorded	Blocking for original-weight outcome validation
F5	Same-close execution and shortened lag sensitivity	Next-close to next-close contract, drift-aware costs, calendar regression tests	Revised implementation; live execution evidence outstanding
F6	Nonpositive/missing prices	Positive input filters; no zero-filling held-name returns; common sample end	Source data limitations remain
F7	Validator's cost-biased permutation comparison	Gross-return comparison, finite-sample p correction	Corrected; conditional-null limitations disclosed
F8	Full-sample tests included development observations	OOT-only Q1 in code and regenerated report	Corrected; reused OOT is not pristine
F9	Limited CLAM experiments generalized to a whole approach	Conclusions limited to tested specifications/data/period	Overstatement withdrawn
F10	Incomplete multiple-testing accounting	Minimum 17-trial registry, 15 available returns, provisional DSR	Open until search history/evidence complete
F11	Direction metric used signs after scaling	Zero-threshold correction, same-prediction audit, corrected new-training monitor	Metric fixed; legacy weights/early selection unchanged
F12	Cross-horizon calibration interpretation	Target-matched diagnostics; ranks excluded; association relabeled	Corrected; no calibration-based approval
F13	Independent-review / exact-reproduction overclaims	Self-review role, model differences and artifact hashes recorded	Disclosure corrected

9. Decisions and resubmission conditions

All GBM/CLAM production uses remain unapproved in this project. clam_orig additionally lacks original artifact/cutoff evidence. Momentum remains only a comparator. No quarterly model family is rejected in general, and no recommendation to abandon all sequence models is supported.

Resubmission requires: frozen intended use and horizon; verified training/data/model provenance; complete experiment registry; point-in-time universe or explicitly bounded bias evidence; realistic execution/cost/capacity validation; matched-target diagnostics; and new untouched observations after specification freeze. Trial counts carry forward. A quarterly-only study would be a different scope requiring quarterly prediction targets and portfolio holding rules, not a rewrite of these weekly-use results.

Weekly monitoring reviews PSI, discrimination, relative drawdown and active SR; a breach or missing evidence suspends the experimental strategy and

triggers investigation. Revalidation is required after material model/data/execution changes. No monitored model is currently authorized for production by this report.

10. Reproduction and evidence

Run `sh scripts/revalidate.sh` against the existing frozen DB and saved scores. It rebuilds the panel, controlled GBM reproduction, portfolio returns, OOT tests, monitoring, gates, report and figures without reselecting weekly models. Run `scripts/audit_clam_metric.py` with TensorFlow separately to rebuild the metric audit. `tests/test_validation_contracts.py` covers timing, missing data, drift, PSI tails, OOT filtering and metric threshold semantics.

The updated notebook workpapers query this same database. CSVs beside this report expose the tables; `review_manifest.json` records code/artifact hashes and the evaluation contract. The PDF is generated from this report. No fresh price download or production deployment was performed. Historical CLAM weights are preserved; the separately registered 500/3000-ticker runs are new trained artifacts.

The review structure is inspired by conceptual-soundness review, ongoing monitoring and outcome analysis in the historical SR 11-7 guidance. This is a methodological reference, not a claim of current regulatory compliance.